Advanced techniques in applied economics

Lecture 1: Conditional independence as structure

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About the course

Why this course?

- We live in a world of high-dimensional dependence.
- Standard tools often break down when too many variables interact.
- Economists increasingly work with data where structure matters: networks, latent factors, strategic interactions, and policy spillovers.

1. Efficiency

Conditional independence (CI) tells us what can be ignored. This can turn an impossible problem into a tractable one.

2. Interpretation

Graphical structure helps separate direct from indirect relations. This is crucial for causal and policy reasoning.

3. Expressivity

Latent variables capture hidden drivers such as confounders, common shocks, market factors, and unobserved heterogeneity.

Core message

Conditional independence + graphs + latent variables = scalable structure for modern empirical work

Course structure and objectives

Format

- **8 content lectures + 2 project sessions**
- Mixture of theory, examples, and hands-on implementation

Tools

- Mathematical derivations and visual intuition
- `R` examples using packages such as `huge`, `glasso`, `pcalg`, `dagitty`

Applied economics angle

I will try to connect the theory to problems economists care about:

- causal identification and policy design,
- financial and macroeconomic dependence,
- latent heterogeneity and confounding,
- interference and network spillovers.

Assessment

The main assessment is a research project. You may choose:

- **Applied project:** empirical analysis of a structured model.
- **Methods project:** deeper study of one model class or algorithm.

Roadmap for today

1. Basic notions: joint, marginal, conditional distributions
2. Independence and conditional independence
3. Regression and the Gaussian bridge
4. Why conditioning can change conclusions
5. Testing dependence in real data

Conditioning is not a technicality — it is often the whole story.

Reading:

S. Højsgaard, D. Edwards, S. Lauritzen, *Graphical Models with R*.
R. Alexander, *Telling stories with data*, Chapter 15.

Part 1: Conditional independence

Random vectors and joint distributions

Let (X, Y) be a random vector.

Joint distribution

It describes the full probabilistic behaviour of (X, Y).

- **Continuous case:** density $f_{XY}(x, y)$
- **Discrete case:** probability mass function

$$f_{XY}(x, y) = \mathbb{P}(X = x, Y = y)$$

A joint distribution describes how variables move *together*, e.g.:

- inflation and unemployment,
- returns on two assets,
- income and consumption,
- treatment assignment and outcomes.

Marginal distributions

Marginals are obtained by summing or integrating out the other variable from $f_{XY}(x, y)$.

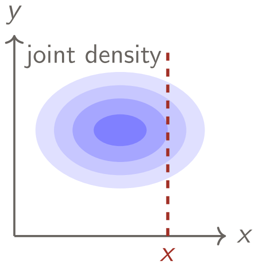
Marginal distribution of X

Continuous case:

$$f_X(x) = \int_{\mathbb{R}} f_{XY}(x, y)\, dy$$

Discrete case:

$$f_X(x) = \sum_y f_{XY}(x, y) = \mathbb{P}(X = x)$$



collect mass in the vertical slice

Conditional distributions

A conditional distribution tells us how one variable behaves once another variable is known.

Conditional distribution

For all x such that $f_X(x) > 0$,

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}.$$

In the discrete case this becomes

$$\mathbb{P}(Y = y \mid X = x) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(X = x)}.$$

Conditioning means that we restrict attention to a subpopulation.

- among employed workers,
- among treated units,
- among firms in distress,
- among borrowers with a given credit score.

Note: Be careful interpreting conditional probabilities

Scenario: AI fraud detection

An algorithm flags fraudulent transactions (F). It catches 90% of fraud, and correctly labels 90% of clean transactions.

Joint probabilities

	Fraud (F)	Clean (F^c)
Flag (+)	0.009	0.099
No flag (-)	0.001	0.891

- **Sensitivity:** $\mathbb{P}(+ \mid F) = 0.9$
- **Specificity:** $\mathbb{P}(- \mid F^c) = 0.9$
- **Prevalence:** $\mathbb{P}(F) = 0.01$

Direction of conditioning matters

If a transaction is flagged, the probability that it is actually fraud equals

$$\mathbb{P}(F \mid +) = \frac{0.009}{0.009 + 0.099} \approx 0.083.$$

So even a strong test may generate many false alarms when the event is rare.

Independence

X and Y are **independent**, written $X \perp\!\!\!\perp Y$, if and only if

$$f_{XY}(x, y) = f_X(x) f_Y(y) \qquad \text{for all } x, y.$$

Equivalent reformulation

$X \perp\!\!\!\perp Y$ if and only if

$$f_{Y|X}(y|x) = f_Y(y) \qquad \text{for all } x \text{ such that } f_X(x) > 0.$$

So learning X does not change the distribution of Y.

Independence means that knowing X does not help to predict Y.

Conditional expectation and mean independence

The conditional expectation of Y given $X = x$ is

$$\mathbb{E}[Y \mid X = x] = \int y \, f_{Y|X}(y \mid x) \, dy.$$

$\mathbb{E}[Y \mid X]$ is the best predictor of Y from X under squared loss (**regression function**).

Mean independence

We say that X and Y are mean independent if $\mathbb{E}[X|Y] = \mathbb{E}[X]$ and $\mathbb{E}[Y|X] = \mathbb{E}[Y]$.

Mean independence is strictly weaker than independence.

Conditional independence

X and Y are **independent given Z**, written $X \perp\!\!\!\perp Y \mid Z$, if

$$f_{XY|Z}(x, y|z) = f_{X|Z}(x|z) f_{Y|Z}(y|z) \qquad \text{for all } x, y, \text{and } z.$$

Equivalently

$$f_{Y|XZ}(y|x, z) = f_{Y|Z}(y|z).$$

Once Z is known, learning X gives no additional information about Y.

Conditional independence captures absence of a direct link.

Why economists should care

Conditional independence is everywhere in applied work.

Causal identification

Unconfoundedness is a conditional independence statement:

$$(Y(1), Y(0)) \perp\!\!\!\perp D \mid X.$$

Selection problems

Apparent differences may disappear or reverse after conditioning on the right variables.

Finance and macro

Sparse dependence means that many variables are unrelated once the right controls are included.

Latent structures

Many models are built by asserting that certain variables become independent once hidden factors are known.

In empirical work, assumptions are often best understood as CI restrictions.

Part 2: Regression and the Gaussian bridge

CI and the regression function

Suppose

$$Y = \beta X + \gamma Z + \varepsilon, \qquad \varepsilon \perp\!\!\!\perp (X, Z).$$

If $\beta = 0$, then

$$Y \perp\!\!\!\perp X \mid Z.$$

So after controlling for Z, the variable X no longer matters.

Caution

If we only assume $\mathbb{E}[\varepsilon \mid X, Z] = 0$, then $\beta = 0$ implies only that the *conditional mean* does not depend on X. This does **not** imply full conditional independence in general.

The multivariate normal distribution

A random vector $X = (X_1, \ldots, X_m)$ is Gaussian, written $X \sim \mathcal{N}_m(\mu, \Sigma)$, if the density is

$$f(x) = \frac{1}{(2\pi)^{m/2}|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right\},$$

where $\mu \in \mathbb{R}^m$ is the mean vector and $\Sigma \in \mathbb{S}^m_+$ is the covariance matrix.

Why this matters

In Gaussian models, many conditional independence questions become algebraic. This is why Gaussian graphical models are so useful.

Gaussian marginals $X \sim \mathcal{N}(\mu, \Sigma)$

Partition X into two subvectors:

$$X = \begin{pmatrix} X_A \\ X_B \end{pmatrix}, \qquad \mu = \begin{pmatrix} \mu_A \\ \mu_B \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{pmatrix}.$$

Marginal law for Gaussians

$$X_A \sim \mathcal{N}(\mu_A, \Sigma_{AA}).$$

To study X_A alone, we simply *drop* the other coordinates. No integration is needed.

Gaussian families are closed under marginalization.

Gaussian conditionals

Under the same partition, the conditional distribution of X_A given $X_B = x_B$ is Gaussian:

$$\mathbb{E}[X_A \mid X_B = x_B] = \mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(x_B - \mu_B),$$
$$\mathrm{var}(X_A \mid X_B = x_B) = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}.$$

Note that:

- The conditional mean is linear in x_B.
- The conditional covariance matrix does not depend on x_B!

Under Gaussianity, conditioning preserves Gaussianity and linearity.

A special Gaussian feature

For general distributions,

$$\mathrm{cov}(X_A, X_B) = 0$$

does **not** imply independence.

For jointly Gaussian vectors, however, we have the equivalence

$$X_A \perp\!\!\!\perp X_B \qquad \Longleftrightarrow \qquad \Sigma_{AB} = 0.$$

Practical implication

Many tools exploit Gaussian logic, even when this is not made explicit.

e.g. some "independence" tests test zero covariance.

The precision matrix

The precision matrix is

$$K = \Sigma^{-1}.$$

Partition it in the same way:

$$K = \begin{pmatrix} K_{AA} & K_{AB} \\ K_{BA} & K_{BB} \end{pmatrix}.$$

Using block matrix inversion and the Schur complement, one obtains

$$\mathrm{var}(X_A \mid X_B = x_B) = K_{AA}^{-1}.$$

Why this is important

- The covariance matrix describes marginal dependence.
- The precision matrix describes conditional dependence.

Precision matrix and conditional independence

Let $X = (X_1, \ldots, X_m)^\top \sim \mathcal{N}_m(\mu, \Sigma)$ and let $K = \Sigma^{-1}$.

For Gaussian data,

$$X_i \perp\!\!\!\perp X_j \mid X_{\{1,\ldots,m\}\setminus\{i,j\}} \qquad \Longleftrightarrow \qquad K_{ij} = 0.$$

Interpretation

A zero off-diagonal entry in the precision matrix means: once all other variables are controlled for, there is no remaining direct relation between X_i and X_j.

This is the bridge to graphs

Later we will encode these zeros by missing edges in an undirected graph.

Partial correlation

For three variables X, Y, Z, the partial correlation is

$$\rho_{X,Y\cdot Z} = \frac{\rho_{X,Y} - \rho_{X,Z}\rho_{Y,Z}}{\sqrt{(1-\rho_{X,Z}^2)(1-\rho_{Y,Z}^2)}}.$$

Meaning

It measures the residual linear association between X and Y after removing the linear effect of Z.

Regression interpretation

$\rho_{X,Y\cdot Z} = 0$ if and only if in the linear regression of X on Y and Z, the coefficient of Y is zero.

In the Gaussian case

Zero partial correlation is equivalent to conditional independence.

Economic interpretation of sparse precision matrices

Suppose X is a vector of many macro or financial variables.

Dense covariance

A dense covariance matrix means many variables move together marginally. This may simply reflect common shocks.

Sparse precision

A sparse precision matrix means that once the rest is controlled for, only a few pairs remain directly related.

Examples

- financial contagion networks,
- interbank or input-output dependence,
- psychometric and preference networks,
- sparse dependence in large macro panels.

Part 3: Why conditioning matters

A big lesson

Simpson's paradox and Berkson's paradox are not curiosities.

Statistical conclusions depend on what we condition on.

- The same data may suggest opposite conclusions under different conditioning sets.
- Conditioning can remove or add dependence.
- Positive association can become negative after conditioning and vice versa.

Conditioning can clarify a relationship, but it can also manufacture one.

An applied economics version of Simpson's paradox

Suppose we compare outcomes across two groups, say A and B.

Naive comparison: We observe

$$\mathbb{E}[Y \mid G = B] > \mathbb{E}[Y \mid G = A].$$

This might tempt us to conclude that group B does better.

But what if the groups differ in skill, education, sector, age, or risk exposure?

Then the marginal comparison may mix:

- composition effects,
- treatment effects,
- selection effects.

Lesson: Before interpreting an association, ask

What should I condition on?

Example 1: Kidney stones

Ignoring confounding can lead to wrong causal conclusions.

- Study of 700 patients with kidney stones [Charig,1986].
- Two treatments:
  - $T = a$: open surgery
  - $T = b$: percutaneous nephrolithotomy
- Overall recovery rates:
  - $\mathbb{P}(R = 1 \mid T = a) = 0.78$
  - $\mathbb{P}(R = 1 \mid T = b) = 0.83$

Based on overall rates, treatment b looks better.

Conditioning on stone size

Patients can be divided into two groups: small/large kidney stones

	Overall	Small stones	Large stones
Treatment *a*	78%	93%	73%
Treatment *b*	83%	87%	69%

Treatment *a* is better in both subgroups.

Stone size acts as a **confounder**:

- Large stones are harder to treat
- More severe cases were more likely assigned to treatment *a*

Example 2: UC Berkeley admissions example

The admission figures of the grad school at UC Berkeley in 1973: 8442 (44%) men, 4321 (35%) women admitted.

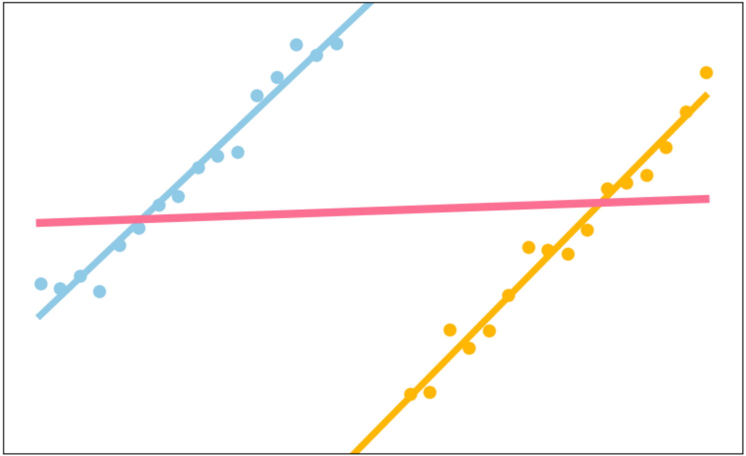
The same data conditioned on the department are:

Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	**82%**
B	560	63%	25	**68%**
C	325	**37%**	593	34%
D	417	33%	375	**35%**
E	191	**28%**	393	24%
F	373	6%	341	**7%**

"Measuring bias is harder than is usually assumed, and the evidence is sometimes contrary to expectation."

(Bickel et al, *Sex Bias in Graduate Admissions: Data From Berkeley*, Science, 1975)

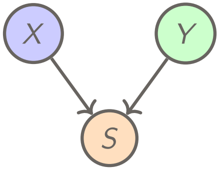
Simpson's paradox: Illustration



Berkson's paradox: conditioning on a collider

Suppose Looks (X) and Personality (Y) are independent in the population.

You only date people who are either attractive or nice. Then being selected into your dating pool depends on both variables.



Lesson

Even if $X \perp\!\!\!\perp Y$ in the population, conditioning on the common effect S may induce dependence:

$$X \not\perp\!\!\!\perp Y \mid S.$$

Economic versions of Berkson's paradox

Selection can create misleading dependence in many economic settings.

Examples

- **Wage and job satisfaction** among workers who stay in a demanding sector.
  Workers with lower wages may stay only if job satisfaction is high, and vice versa.
- **Ability and family background** among students admitted to a selective program.
  Students with weaker background may enter only if ability is very high, and vice versa.

General lesson

These variables may be weakly related, or even unrelated, in the full population. But after conditioning on the selection event, they can become associated.

Conditioning on a consequence can create dependence that was not there before.

Exercise: mediation

Suppose

$$X,\ \varepsilon_Z,\ \varepsilon_Y \quad \text{are jointly independent,}$$

and

$$Z = aX + \varepsilon_Z, \qquad Y = bZ + \varepsilon_Y.$$

1. Is $X \perp\!\!\!\perp Y$?
2. Show that $Y \perp\!\!\!\perp X \mid Z$.
3. How would a regression of Y on X alone be interpreted?

Economic reading

X could be a policy variable, Z an intermediate behavioural response, and Y an outcome.

Exercise: confounding

Suppose

$$X = aU + \varepsilon_X, \qquad Y = bU + \varepsilon_Y,$$

with independent noise terms.

1. Is $X \perp\!\!\!\perp Y$?
2. Show that $X \perp\!\!\!\perp Y \mid U$.
3. Interpret U as an omitted variable.

Economic reading

U could be ability, firm quality, local demand, or risk preferences.

Exercise: selection bias

Suppose

$$X \perp\!\!\!\perp Y \qquad \text{and} \qquad S = X + Y + \varepsilon.$$

1. Are X and Y independent in the population?
2. What may happen to $\mathrm{cor}(X, Y \mid S > 0)$?
3. Give a real empirical example.

Hint
Selection is often an implicit conditioning step.

Part 4: Testing dependence in data

From theory to data

So far we discussed population properties such as

$$X \perp\!\!\!\perp Y \qquad \text{or} \qquad X \perp\!\!\!\perp Y \mid Z.$$

In data, we only observe a sample:

$$(X^{(1)}, Y^{(1)}), \ldots, (X^{(n)}, Y^{(n)}).$$

Goal: Use the sample to decide whether the dependence we see is plausibly just noise.

Practical question: Which notion of dependence are we testing?

- linear dependence,
- arbitrary dependence.

A first test: Pearson correlation

The most familiar test is based on the sample correlation coefficient r_n.

Consider the test $$H_0 : \rho = 0 \qquad \text{vs.} \qquad H_A : \rho \neq 0.$$

When does this make sense?

Mainly when dependence is approximately linear and the Gaussian approximation is reasonable.

This is a test for **vanishing linear association**, not for full independence in general.

Zero correlation is not the same as independence.

Fisher's z-transform test

Let r_n be the sample correlation from an *iid* sample $(X^{(1)}, Y^{(1)}), \ldots, (X^{(n)}, Y^{(n)})$. Define

$$Z_n = \frac{1}{2} \log\left(\frac{1 + r_n}{1 - r_n}\right).$$

If (X, Y) is bivariate normal with correlation ρ, then asymptotically

$$Z_n \approx \mathcal{N}\left(\frac{1}{2} \log\left(\frac{1 + \rho}{1 - \rho}\right), \frac{1}{n - 3}\right).$$

```r
# simulate sample correlations
library(MASS)
set.seed(1)
n <- 500
iter <- 1000
rho <- 0.2
srho <- rep(0, iter)
Sigma <- matrix(c(1, rho, rho, 1), 2, 2)
for (i in 1:iter) {
  x <- mvrnorm(n, mu = c(0, 0), Sigma = Sigma)
  srho[i] <- cor(x[,1], x[,2])}
```

```r
# Fisher transform and comparison
zrho <- 0.5 * log((1 + srho)/(1 - srho))

hist(zrho, prob = TRUE, breaks = 30,
    main = "", xlab = "z")

# we compare the result with the theoretical limit
curve(dnorm(x,
      mean = 0.5*log((1+rho)/(1-rho)),
      sd = 1/sqrt(n-3)),
      add = TRUE, lwd = 2)
```

Pearson test in R

Fisher's test is implemented in R through `cor.test()`.

```
library(MASS)
set.seed(1)
rho <- 0.2
Sigma <- matrix(c(1, rho, rho, 1), 2, 2)
x <- mvrnorm(n, mu = c(0, 0), Sigma = Sigma)

cor.test(x[,1], x[,2], method = "pearson")
```

Try the same with $\rho = 0$.

What does the *p*-value mean?

It measures how surprising the observed sample correlation would be under the null hypothesis.

But remember

A non-significant result does **not** imply independence.

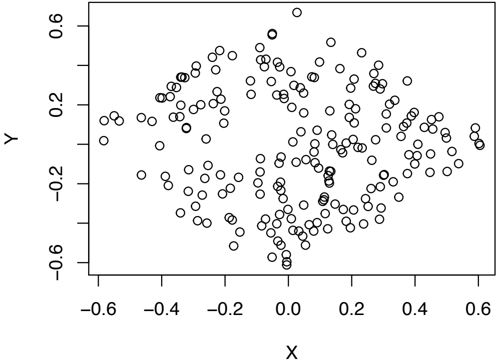
Limitation of Pearson correlation

A pair of variables can be strongly dependent and still have zero correlation.

Why? Correlation only measures one aspect of dependence: the linear component.

Examples

- nonlinear deterministic relationships,
- symmetric patterns,
- dependence driven by tails or selection.



Implication

If economic data are heavy-tailed or nonlinear, Pearson correlation can miss important structure.

Non-Gaussianity issue

Vanishing covariance does not imply independence!

```
# generate sample from two uncorrelated but dependent random variables
> set.seed(1); n <- 200
> A <- runif(n)-1/2; B <- runif(n)-1/2
> X <- t(c(cos(pi/4),-sin(pi/4)) %*% rbind(A,B))
> Y <- t(c(sin(pi/4),cos(pi/4)) %*% rbind(A,B))
> cor.test(X,Y, method = "pearson")

^^IPearson's product-moment correlation

data: X and Y
t = -0.84711, df = 198, p-value = 0.398
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.1971897 0.0793095
sample estimates:
       cor
-0.06009275
```

X and Y are uncorrelated but dependent!

Distance correlation

Distance correlation is a fully nonparametric measure of dependence.

Key property: $\mathcal{R}(X, Y) = 0 \iff X \perp\!\!\!\perp Y.$

```
library(energy)
set.seed(1)
n <- 200
A <- runif(n) - 1/2
B <- runif(n) - 1/2
X <- A + B
Y <- A^2 - B^2

dcor.test(X, Y, R = 1000)
```

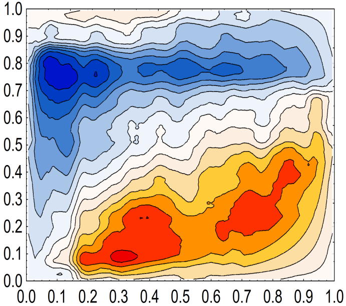
Interpretation

This gives a permutation test for general dependence, not just linear or monotone association.

Another cautionary example

Bowman& Azzalini (1997) analyse aircraft wing span and speed data.

```
> library(sm); set.seed(1);
> X <- aircraft$Span
> Y <- aircraft$Speed
> cor.test(X,Y)$p.value
[1] 0.7816014
> dcor.test(X,Y,R=1000)$p.value
[1] 0.000999001
```



For a discussion see: Ćmiel, Ledwina, *Validation of Association*, 2019.

χ^2-test for discrete data

```
> M <- as.table(rbind(c(762, 327, 468), c(484, 239, 477)))
> dimnames(M) <- list(gender = c("F", "M"),
+ party = c("Democrat","Independent", "Republican"))

> (Xsq <- chisq.test(M)) # Prints test summary

^^IPearsons Chi-squared test

data: M
X-squared = 30.07, df = 2, p-value = 2.954e-07

> Xsq$expected # expected counts under the null
      party
gender Democrat Independent Republican
     F 703.6714 319.6453 533.6834
     M 542.3286 246.3547 411.3166
```

df = 2 is the difference between 5 (saturated model) and 3 (independence)

Testing conditional independence

Testing $X \perp\!\!\!\perp Y \mid Z$ is much harder than testing ordinary independence. Once we condition on Z, we must compare *conditional* laws rather than marginal ones.

- **Discrete data:** likelihood-ratio gives asymptotic χ^2 procedures.
- **Gaussian data:** conditional independence reduces to zero partial correlation.
- **Nonlinear / non-Gaussian data:** kernel and other nonparametric methods are available, but they are computationally heavier.

```r
library(CondIndTests)
library(bnlearn)
set.seed(1)

n <- 100
Z <- rnorm(n)
X <- 4 + 2*Z + rnorm(n)
Y <- 3*X^2 + Z + rnorm(n)

CondIndTest(X, Y, Z,
            method = "KCI")$pvalue
bnlearn::ci.test(X, Y, Z)$p.value
```

In practice, the difficulty of CI testing depends strongly on the modelling assumptions.

Interpretation

Here X and Y are **not** conditionally independent given Z, so both procedures return very small p-values.

Testing conditional independence in discrete data

Using bnlearn

```
library(gRim)
data(UCBAdmissions)

bnlearn::ci.test(
  x = "Gender",
  y = "Admit",
  z = "Dept",
  test = "x2",
  data = as.data.frame(UCBAdmissions)
)
```

```
Pearson's X^2

data: Gender ~ Admit | Dept
x2 = 0, df = 6, p-value = 1
alternative hypothesis:
true value is greater than 0
```

Using gRim

```
gRim::ciTest(
  as.data.frame(UCBAdmissions),
  set = ~ Gender + Admit + Dept
)
```

```
Testing Gender _|_ Admit | Dept
Statistic (DEV): 0.000
df: 6
p-value: 1.0000
method: CHISQ

Slice information:
  statistic p.value df Dept
1 0 1 1 A
2 0 1 1 B
3 0 1 1 C
4 0 1 1 D
5 0 1 1 E
6 0 1 1 F
```

Take-away

- Independence means factorization of the joint law.
- Conditional independence is a structural statement about what remains after conditioning.
- Conditioning can remove dependence, but it can also create it.
- In Gaussian models, conditional independence becomes algebraic: zeros in the precision matrix correspond to missing direct relations.
- Testing dependence requires care: Pearson and distance correlation target different notions.

Main lesson of today: to understand dependence, always ask what is being conditioned on.

Looking ahead

Next lecture

We will encode conditional independence statements visually using **undirected graphs**.

What will come next?

- local, pairwise, and global Markov properties,
- Gaussian graphical models,
- sparse network estimation from data,
- applications to economic and financial dependence networks.

$$\text{Missing edge} = \text{conditional independence}$$